

TECHNICAL INSPECTION REPORT: ATMOSPHERIC COLUMN V-701A

Asset ID: NRC-REF-V-701A | Service: Crude Distillation | Inspection Cycle: Q3 Annual Integrity

1. Operating Environment & Design Specifications

Parameter	Design Rating	Current Operating Point	Tolerance Limit
Operating Pressure	45.0 PSIG	39.4 PSIG	42.0 PSIG
Operating Temperature	365.0 °C	358.2 °C	380.0 °C
Design Corrosion Allowance	3.00 mm	2.85 mm consumed	Exceeded in Lower Trays
Primary Metallurgy	ASTM A516 Gr 70 Carbon Steel	Shell Clad 410S Stainless	ASME Section VIII Div 1

2. Ultrasonic Wall Thickness Gauging (NDT Findings)

Inspection Location	Nominal (mm)	Minimum Retirement (mm)	Measured Actual (mm)	Status Verdict
Top Dome Section (Elevation +42m)	10.00	7.50	9.40	COMPLIANT
Shell Course 1 (Trays 28-35)	12.00	9.50	11.10	COMPLIANT
Shell Course 2 (Trays 15-27)	14.00	11.00	12.30	MONITOR
Shell Course 3 (Flash Zone Elevation +12m)	16.00	12.80	10.45	CRITICAL THINNING
Bottom Sump & Reboiler Return	18.00	14.00	15.90	COMPLIANT

3. Auxiliary Relief Devices & Defect Observations

- **Pressure Relief Valve PRV-701A:** Tested on bench. Valve exhibited seat leakage at 36 PSIG (below 45 PSIG set pressure) and reseal delay of 4.8 seconds during full lift simulation.
- **Tray 18 Hold-down Clamps:** Three localized clips detached due to severe sulfide stress cracking (SSC). Debris lodged near downcomer nozzle.
- **Flash Zone Internal Cladding:** 340mm area of 410S stainless cladding has delaminated, exposing base carbon steel shell to naphthenic acid attack at 345°C.

4. Inspector Conclusions & Required Action

Due to Shell Course 3 wall thickness measuring 10.45 mm against the minimum statutory retirement threshold of 12.80 mm, immediate regulatory notification and mechanical reinforcement or weld overlay is mandatory before column re-commissioning.

Inspected By: K. Deshmukh, NDT Level-III Certification ID #88412 | Verified: Chief Plant Inspector